

Software Quality Assurance (SQA) Plan

By SC3040 TEL5 Team 2

Dylan Quek Zhi En
Ang Wei Ming
Chi Seo Hyeon
Choo Kean Yee
Madelyn Cruz Tan
Lim Jun Howe
Tan Yong Huat

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Signature Page

Prepared by: Yong Huat Date: 19/9/2025

Reviewed by 1: Kean Yee Date: 19/9/2025

Reviewed by 2: Jun Howe Date: 19/9/2025

Approved by : Seohyeon Date: _____

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1. Purpose and Scope

1.1. Purpose

The purpose of this Software Quality Assurance (SQA) Plan is to establish the goals, processes, and responsibilities required to implement effective quality assurance functions for the ReportNow project.

The Software Quality Assurance Plan provides the framework necessary to ensure a consistent approach to software quality assurance throughout the project life cycle. It defines the approach that will be used by the QAM and Software Quality (SQ) personnel to monitor and assess software development processes and products to provide objective insight into the maturity and quality of the software. The systematic monitoring of products, processes, and services will be evaluated to ensure they meet requirements and comply with policies, standards, and procedures, as well as applicable Institute of Electrical and Electronic Engineers (IEEE) and ISO standards.

1.2. Scope

The purpose of SQA is to ensure that the software developed does not deviate from the original intended product. SQA is also concerned to identify any errors, omissions, inconsistencies, and alternatives, enhancements or improvements that can be made at any stage of development.

This SQA Plan covers the design, implementation, testing, and deployment of *ReportNow*: a web/mobile platform that lets citizens submit and track public-issue reports which are routed to the appropriate agencies. It applies to the Citizen, Government, and Admin user interfaces and their supporting MVC services and models, including role-based access. It also covers AI-assisted issue categorization and duplicate detection features. Included components are the Next.js/Node.js services, Prisma/PostgreSQL persistence, authentication and APIs used by these UIs

2. Reference Documents

- IEEE STD 730-2002, IEEE Standard for Software Quality Assurance Plans (http://standards.ieee.org/reading/ieee/std_public/description/se/730-2002_desc.html)
- ISO IEC 90003:2004 Software Standard (<http://praxiom.com/iso-90003.htm>)
- Project Proposal
- System Requirement Specifications

[Update the reference document list to include a list of project documents used or referenced in the development of this plan. This includes policies, standards, procedures, guidelines, and other similar documents. Note: the last documents listed are examples of project documents that you might include.]

3. Management

This section describes the management organizational structure, its roles and responsibilities, and the software quality tasks to be performed.

3.1. Management Organisation

The implementation of the quality assurance system is the responsibility of the Quality Assurance Manager (QAM).

3.1.1. Project Management

The Project Manager will be responsible for approving:-

- The system requirement specification document
- The overall time scale for the project
- The choice of system development life cycle
- The choice of software development tools and techniques utilised
- The selection of project teams
- The training of project teams

3.1.2. Assurance Management

The QAM provides Project Management with visibility into the processes being used by the software development teams and the quality of the products being built. The QAM maintains a level of independence from the project and the software developers.

In support of software quality assurance activities, the QAM has assigned and secured Software Quality personnel from the pool of available SQ trainees to coordinate and conduct the SQ activities for the project and report back results and issues.

3.2. Tasks

This section summarizes the tasks (product and process assessments) to be performed during the development of software. These tasks are selected based on the developer's Project Plan and planned deliverables, and identified reviews.

- Backend & Core Services

- Set up backend environment and database (PostgreSQL with Prisma).
- Implement API routes for authentication, issue management, and notifications.

- User Management

- User registration

- User login and session management.
- Manage user account (update details, deactivate/reactivate).
- **Issue Reporting & Viewing**
 - Submit and manage reports (title, description, location, image).
 - View reported issues (with images and metadata).
 - Filter and search issues by category, date, or location.
- **AI-Enabled Functions**
 - Categorize and assign issues using AI ($\geq 80\%$ accuracy).
 - Detect duplicate reports within seconds of submission.
- **Issue Management**
 - Verify and update issue status (open, in-progress, resolved).
 - Reassign issues to appropriate agencies.
 - Mark issues as done.
- **Community Features**
 - Upvote reported issues (crowdsourced prioritization).
 - Leave and view reviews/ratings for issue resolutions.
 - Notifications (general updates and opt-in alerts).

3.2.1. Product Assessments

The following product assessments will be conducted by SQ personnel:

- **Citizen User Interface (CitizenUI):** Verify usability, responsiveness, and accessibility of the issue reporting and tracking functions.
- **Government User Interface (GovUI):** Ensure reliability of issue assignment, status updates, and communication with citizens.
- **Admin User Interface (AdminUI):** Validate ability to reclassify issues, manage accounts, and oversee system performance.
- **Authentication & Security Module:** Assess Singpass login, registration guardrails, and role-based access control for compliance with security

requirements.

- **AI Categorization & Duplicate Detection:** Test classification accuracy ($\geq 80\%$) and duplicate flagging speed (< 5 seconds) against benchmark datasets.
- **Backend APIs & Database (Node.js + PostgreSQL):** Evaluate correctness of API endpoints, data integrity in issue logs, and response time under load.

3.2.2. Process Assessments

The following process assessments will be conducted by SQ personnel:

- **Requirements Management Process:** Evaluate completeness and traceability of functional and non-functional requirements across development stages.
- **Configuration & Version Control Process:** Confirm consistent use of GitHub for code integration, branch management, and release tagging.
- **Change Management Process:** Review handling of feature updates, bug fixes, and scope changes, including approval workflows.
- **Testing & Validation Process:** Assess unit, integration, and system testing coverage, including automated tests with Jest and React Testing Library.
- **Code Review & Quality Assurance Process:** Verify adherence to coding standards, documentation quality, and peer review practices.
- **Deployment & Release Process:** Ensure production readiness through deployment validation, rollback procedures, and monitoring setup.

3.3. Roles and Responsibilities

This section describes the roles and responsibilities for each assurance person assigned to the Project.

3.3.1. QAM

Responsibilities include, but are not limited to:

- Secure and manage SQ personnel resource levels
- Ensure that SQ personnel have office space and the appropriate tools to conduct SQ activities
- Provide general guidance and direction to the SQ personnel responsible for conducting software quality activities and assessments

- Assist SQ personnel in the resolution of any issues/concerns and/or risks identified as a result of software quality activities
- Escalate any issues/concerns/risks to project management

3.3.2. Software Quality Personnel

Responsibilities include, but are not limited to:

- Develop and maintain the project software quality assurance plan
- Generate and maintain a schedule of software quality assurance activities
- Conduct process and product assessments, as described within this plan
- Identify/report findings, observations, and risks from all software assurance related activities to the QAM

4. Documents

4.1. Purpose

This section identifies the minimum documentation governing the requirements, development, verification, validation, and maintenance of software that falls within the scope of this software quality plan. Each document below shall be assessed (reviewed) by SQ personnel.

4.2. Minimum Document Requirements

- Backlog
- Meeting minutes
- Project proposal
- Use case model
- System requirement specification
- Quality Plan
- Project Plan
- Risk Management
- Prototype related items, such as code, documents, slides, video clips, etc.
- Design report on software maintainability
- Configuration Management Plan
- Change management plan
- Release plan
- Test Plan
- Test Cases and Requirements Test Coverage report

5. Standards, Practices, Conventions and Metrics

5.1. Purpose

This section highlights the standards, practices, quality requirements, and metrics to be applied to ensure the successful completion of the software quality programme.

5.2. Software Quality Programme

These practices and conventions are tools used to ensure a consistent approach to software quality for all programs/projects.

The four aspects of the software quality programme that are deemed to have the most significant effect on the product are as follows:

1. **Usability** – The product must be intuitive and accessible to all users, regardless of technical background, ensuring that issues can be reported quickly without confusion.
2. **Reliability** – User submissions must be submitted, stored, and tracked without data loss or system crashes, even under high user loads.
3. **Maintainability** – The app should be easy to update (e.g., adding new categories of issues or updating forms) without disrupting service.
4. **Consistency** – Both the user interface and reporting workflow must remain uniform across different platforms (mobile and desktop), ensuring a predictable and reliable experience to users.

5.2.1. Standard Metrics

The following standard metrics are the minimum planned metrics that will be collected, reported, and maintained in the area of software quality assurance:

- **App crash rate** - Frequency of unexpected crashes per a set number of sessions.
- **Average report submission time** - Time taken for a user to complete and submit an issue report.
- **Response time** - Average time between a user submitting a report and receiving confirmation/acknowledgement.
- **Defect density in core functions** - Number of defects per KLOC, or per functional module (i.e. submission, geolocation, etc.)
- **Data consistency** - Percentage of reports that appear correctly/incorrectly in both user and admin views without data discrepancies.
- **Error rate in report processing** - Percentage of submissions that fail due to system errors.
- **Submission tracking accuracy** - Percentage of submissions correctly tracked from the users.
- **Usability** - Percentage of test users that can complete core tasks of the software (completing and submitting a report) without guidance.

6. Software Reviews

6.1. Purpose

This section identifies the number and type of system/subsystem reviews and engineering peer reviews that will be supported by the SQ Personnel. The project milestone chart, and the SQ Personnel resource levels determine the reviews that are supported.

6.2. Minimum Software Reviews

For each review, SQ will assess the review products to assure that review packages are being developed according to the specified criteria, the review content is complete, accurate, and of sufficient detail, and Requests for Action are captured, reviewed, and tracked to closure. In addition, SQ will assess the processes used to conduct the reviews to determine if appropriate personnel are in attendance, correct information is presented, entry and exit criteria are met, and appropriate documents are identified for update.

The following software reviews will be assessed by SQ:

- Project Plan Review
- Requirements Analysis Review
- Software Design Review
- Integration Review
- Test Plan and Test Case Review
- Usability Review
- Acceptance Review

7. Test

The Software Quality (SQ) team, led by QA Manager, Yong Huat , and supported by the QA Engineer, Kean Yee, will ensure that all testing processes are implemented according to the test plan. SQ personnel will monitor all testing activities to ensure they adhere to the project timeline as outlined in the Gantt chart.

SQ will verify that all tests are conducted using the approved test frameworks, such as Jest and React Testing Library, and that any anomalies are identified, documented in the issue tracker, and tracked to closure. Additionally, the SQ team will ensure that test results are accurately recorded to validate that the system meets its target specifications.

Some SQ test activities include:

- The AI model's classification accuracy is at least 80% on a labeled test dataset.
- The duplicate detection mechanism flags similar reports within five seconds of submission.

- User registration guardrails reject at least 95% of invalid inputs.
- New reports are visible on the map within three minutes of a successful submission.
- Navigation allows users to access detailed report information within three clicks from the home screen.

Following test execution, the SQ team will review test reports, problem reports, and final test results to ensure all requirements have been met before the project is pushed to production.

8. Problem Reporting and Corrective Action

All findings, bugs, and other issues identified by SQ personnel will be generated, tracked, and managed in a centralized system using GitHub Issues within the project's GitHub repository. This system will act as the single source of truth for all identified problems.

When a problem is found, the QA Engineer will create a new issue with a descriptive title. Each report will include:

- Steps to reproduce the issue.
- The expected outcome and the actual result.
- Relevant information, such as screenshots, error messages, or logs.
- Assignment to the appropriate developer (eg. Dylan for front-end, Seo Hyeon for back-end).

The QA Manager will communicate the status of high-priority issues and overall bug trends to the Project Manager, Madelyn, and the development team during scheduled meetings. This ensures that corrective actions are prioritized effectively to align with the project timeline.

9. Tools, Techniques and Methodologies

SQ personnel will use the following tools and platforms:

- Testing Frameworks:
 - Jest and React Testing Library for automated unit and integration testing of the application's functionality
- Version Control and Issue Tracking:
 - GitHub for version control
 - GitHub Issues for tracking bugs and issues
- Security and Dependency Scanning:
 - Dependabot and Scantist to monitor for vulnerabilities in project dependencies, with alerts that will be reviewed by the SQ team
- Project Management and Documentation:
 - Trello board will be used to track backlog and progress
 - Google Drive will be used for collaboration of project-related documentation, including project proposal and SRS.
- Development and Deployment Platforms:
 - Access to the development and staging environments deployed on Vercel to perform testing

10. Media Control

SQ deliverables will be documented in one of the following Microsoft software applications: Word, Excel, or PowerPoint. Deliverables will be in soft copy, with the exception of completed checklists from process and product assessments. See Section 12 for additional details on the collection and retention of key records. Software Quality personnel will request space on the project's secured server for SQ records. This server is password protected and backed up nightly.

Our project will be stored using the following services

- Google Drive
- GitHub
- Trello

Google Drive is used as the team's main workspace for live collaboration. Drafts, working documents, and in-progress materials are developed here first before being transferred to GitHub once they are cleaned up and finalized. This allows simultaneous editing and real-time feedback among team members.

GitHub serves as our primary platform for version control and collaborative coding. It is used to store tidied and completed work, ensuring that finalized project artifacts are well-documented, traceable, and easy to share among team members.

Trello is our project management tool for tracking tasks, managing backlogs, and organizing workflows. It helps visualize progress through boards and cards, allowing the team to prioritize work, assign responsibilities, and monitor deadlines effectively.

11. Supplier Control

[Not applicable for this project]

12. Record Collection, Maintenance, and Retention

SQ personnel will maintain records that document assessments performed on the project. Maintaining these records will provide objective evidence and traceability of assessments performed throughout the project's life cycle. There are two types of records that will be maintained: Hardcopy and Electronic. SQ personnel will maintain electronic or hard copies of all assessment reports and findings. SQ Project folders will contain hardcopies of the assessment work products such as completed checklists, supporting objective evidence, and notes.

The table below identifies the record types that will be collected, as well as the Record Custodian and Retention period

Record Title	Record Custodian	Record Retention
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SQA Assessments	SQ Personnel	One Year
SQA Checklists	SQ Personnel	One Year
Deliverable Defects	SQ Personnel	One Year

13. Training

SQ personnel have fundamental knowledge in the following areas through prior experience, training, or certification in methodologies, processes, and standards:

- Audits and Reviews (Assessments)
- Risk Management
- Software Assurance
- Configuration Management
- Software Engineering
- ISO 9001, ISO 9000-3
- CMMI
- Verification and Validation
- Prisma ORM (Prisma.io) usage and schema management
- Database migrations and version control using Prisma Migrate

14. Risk Management

SQ personnel will assess the project's risk management process and participate in weekly risk management meetings and report any software risks to the QAM and the project manager.

Risk management plan:

- Monitoring schema changes and ensuring Prisma migrations are peer-reviewed before deployment.
- Identifying risks in database access patterns, such as inefficient queries or improper use of relations, and reporting them to the QAM and the project manager.

- Assessing potential risks related to data integrity, concurrency issues, or rollback procedures.
- Reporting database-related risks or Prisma toolchain issues.

15. SQA Plan Change Procedure and History

SQ personnel are responsible for the maintenance of this plan, including Prisma database support activities. It is expected that this plan will be updated throughout the life cycle to reflect any changes in:

- Support levels
- Prisma ORM features or upgrades
- Software Quality (SQ) activities tied to database assurance

Proposed changes shall be submitted to the Quality Assurance Manager (QAM), along with supportive material justifying the proposed change. For database-related changes, justification should include:

- Schema migration history
- Prisma migration scripts
- Impact assessment on current and future application modules